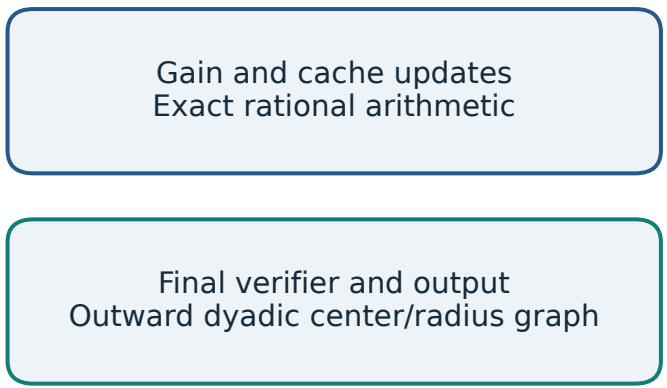
Arithmetic boundaries belong in the contract

Exact cached updates and rounded parameter centers carry different obligations.



Hardware floating point requires
a separate execution refinement.

A covariance-center warning

Exact PSD covariance
[[1/4, 3/4], [3/4, 9/4]]

Down-rounded centers at p = 1
[[0, 1/2], [1/2, 2]]

$v = (-3, 1)$: $v^{T} C v = -1$

Exact arithmetic contract and independent covariance-center counterexample.